

Mitchell C. Bosley

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Research Profile

I am a political scientist and artificial intelligence researcher studying how machine learning and large language models reshape democratic institutions, public policy, and political behavior. My research program is driven by the conviction that artificial intelligence will transform how humans acquire and interpret political information and deserves to be studied as a political phenomenon in its own right. To this end, my work focuses on three areas: (1) developing AI-driven measurement techniques to analyze political institutions and political discourse; (2) understanding how AI-human interaction shapes political attitudes and behavior; and (3) using AI-powered simulations to gain insight into political processes. My work on AI and politics has been published in venues such as the American Political Science Review and the Journal of Political Institutions and Political Economy, with ongoing work on AI-mediated prejudice reduction accepted at PNAS Nexus.

Academic Appointments

Postdoctoral Researcher, CENIA - National Center for Artificial Intelligence
Santiago, Chile, 2025 - Present

Postdoctoral Researcher, Schwartz Reisman Institute for Technology and Society
University of Toronto, 2024 - 2025

Education

University of Michigan

Ph.D., Political Science, 2024 (awarded September 2024)

Fields: *Political Methodology, Comparative Politics, Political Economy*

Dissertation: *Three Papers in the Applied Use of Machine Learning and Artificial Intelligence Models for the Analysis of Political Text Data*

Committee: Walter Mebane (Co-Chair), George Tsebelis (Co-Chair), Yuki Shiraito, Ken Kollman, Charles Shipan

University of British Columbia

M.A., Political Science, 2018

B.A., Political Science (Honors Program), 2016

Publications and Works in Progress

Peer-Reviewed Publications

- *Improving Probabilistic Models in Text Classification via Active Learning*
American Political Science Review
with Saki Kuzushima, Yuki Shiraito, and Ted Enamorado
- *Towards Qualitative Measurement at Scale: A Prompt-Engineering Framework for Large-Scale Analysis of Deliberative Quality in Parliamentary Debates*
Journal of Political Institutions and Political Economy (2025)

Accepted

- *Can AI Help Reduce Prejudice? Evaluating the Effectiveness of AI-Powered Personalized Persuasion on Support for Transgender Rights*
PNAS Nexus (Accepted)
with John Holbein, Charles Crabtree, and Semra Sevi

Works in Progress

1. Text Analysis and Legislative Research

- *Do we still need BERT in the age of GPT? Comparing Domain-Adaptation and In-Context-Learning for Political Text Analysis*
with Musashi Hinck, Hauke Licht, and Alexander Hoyle
- *Mapping Committee Deliberation to Legislative Outcomes: Using Generative AI to Trace Policy Evolution in the U.S. Congress*
with Julia Park and Kevin Esterling
- *Institutional Change After Franchise Expansion: Evidence From British India, 1919-1947*
with Thiha Zaw and Ajit Phadnis

2. AI-Powered Persuasion

- *Reducing Prejudice at Scale: Personalized Persuasion on the Basis of MFQ Placement with AI Chatbots*
with John Holbein, Charles Crabtree, and Semra Sevi

3. Political Behavior Simulation

- *Using Generative AI to Simulate 1000 Canadian Voters*
with Sarah Lachance and Peter Loewen

4. AI Literacy and Development

- *AI Literacy as a Development Tool: A Field Experiment in Peru*
with Charles Crabtree and Veronica Hurtado
- *Can AI Tools Enhance Educational Support? A Study of Parent-Child Learning Interventions*
with Jonathan Collins

Teaching Experience

As Instructor of Record

University of British Columbia

Investigating Politics: An Introduction to Scientific Political Science (2023)
Full responsibility for course design, lectures, assignments, and grading

As Graduate Student Instructor

University of Michigan

Introduction to Game Theory (2020)
Research Ethics for Graduate Students (2020)
Introduction to World Politics (2019)

University of British Columbia

Introduction to Comparative Politics (2017)
Introduction to Game Theory (2017)
Introduction to Canadian Politics (2016)

Awards

- **Rackham Doctoral Fellowship**, University of Michigan
 - 2024: \$3,000 USD
- **Department of Political Science Director of Graduate Studies Award**, University of Michigan
 - 2023: \$1,500 USD
- **Gerald R. Ford Dissertation Fellowship**, University of Michigan
 - 2023-2024: \$25,980 USD
 - 2022-2023: \$24,054 USD
- **Text as Data Association Conference Attendance Grant**
 - University of Massachusetts, Amherst, 2023: \$530 USD
 - Cornell Tech, 2022: \$1,600 USD

- **Empirical Implications of Theoretical Models (EITM) Group Attendance Grant**, University of California, Berkeley
 - 2022: \$900 USD
- **MapleMeth (Canadian PolMeth) Attendance Grant**, University of Western Ontario
 - 2023: \$800 CAD
 - 2022: \$1,100 CAD
- **Weiser Center for Emerging Democracies Research Grant**, University of Michigan
 - 2021: \$4,000 USD
- **Summer Collaboration Grant**, University of Michigan
 - 2021: \$3,000 USD
- **Rackham Doctoral Fellowship**, University of Michigan
 - 2018-2019: \$24,000 USD
- **Rackham MPSA Travel Fund**, University of Michigan
 - 2023: \$500 USD
 - 2022: \$500 USD
- **Conference Travel Fund**, University of Michigan
 - PolMeth Conference, 2023: \$600 USD
 - American Political Science Association, 2023: \$600 USD
 - American Political Science Association, 2022: \$600 USD
 - Canadian Political Science Association, 2019: \$800 USD

Conference Presentations

American Political Science Association

- 2024: *Towards Qualitative Measurement at Scale: A DQI Measurement Framework using Prompt-Engineering*
- 2023: *When do the Rules Matter? Analyzing the Effect of Rule Change on Obstruction in Legislatures*
- 2022: *When do the Rules Matter? Analyzing the Effect of Rule Change on Obstruction in Legislatures* (Poster)
- 2020: *Improving Probabilistic Models in Text Classification via Active Learning* with Saki Kuzushima, Yuki Shiraito, and Ted Enamorado

Society for Political Methodology (PolMeth)

- 2024 (UC Riverside): *Towards Qualitative Measurement at Scale: A DQI Measurement Framework using Prompt-Engineering* (Poster)
- 2023: *Improving Probabilistic Models in Text Classification via Active Learning* with Saki Kuzushima, Yuki Shiraito, and Ted Enamorado

JPIPE Conference (AI for the Study of Political Institutions)

- 2024: *Towards Qualitative Measurement at Scale: A DQI Measurement Framework using Prompt-Engineering* (Paper)

Toronto-Montreal Political Behavior Workshop (McGill University)

- 2024: *Reducing Prejudice at Scale: Personalized Persuasion with Chatbots*

Latin American PolMeth (Universidad de la República, Montevideo)

- 2024: *Reducing Prejudice at Scale: Personalized Persuasion with Chatbots*

Canadian Political Science Association

- 2023: *Using Language Models to Measure Legislative Obstruction in the Canadian House of Commons*
- 2019: *Rationalizing Legislative Rule Change: Three Cases of Rule Change in the Canadian House of Commons*

Canadian Political Methodology Society (MapleMeth)

- 2023: *Using Language Models to Measure Legislative Obstruction in the Canadian House of Commons*
- 2022: *Towards LegisBERT: Improving Language Models for the Analysis of Legislative Speech*

Midwestern Political Science Association

- 2023: *Do we still need BERT in the age of GPT? Comparing the benefits of domain-adaptation and in-context-learning approaches to using LLMs for Political Science Research* with Musashi Jacobs-Harukawa, Hauke Licht, and Alexander Hoyle
- 2022: *Institutional Change After Franchise Expansion: Evidence From British India* with Thiha Zaw
- 2022: *When do the Rules Matter? Analyzing the Effect of Rule Change on Obstruction in Legislatures.*
- 2021: *Improving Probabilistic Models in Text Classification via Active Learning* with Saki Kuzushima, Yuki Shiraito, and Ted Enamorado

Text as Data Association Conference

- 2023: *Measuring Obstructionary Speech in Legislatures using Large Language Models* (Poster)
- 2022: *Towards LegisBERT: Improving Language Models for the Analysis of Legislative Speech* (Poster)
- 2021: *Improving Probabilistic Models in Text Classification via Active Learning* with Saki Kuzushima, Yuki Shiraito, and Ted Enamorado

Research Methods

Quantitative Methods

Machine Learning, Natural Language Processing, Text Analysis, Causal Inference, Survey Experiments

Programming and Computation

R, Python, High Performance Computing (Slurm), Version Control (Git)

Languages

English (Native), French (Intermediate), Spanish (Intermediate)

References

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Walter Mebane, Ph.D.

Professor of Political Science
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